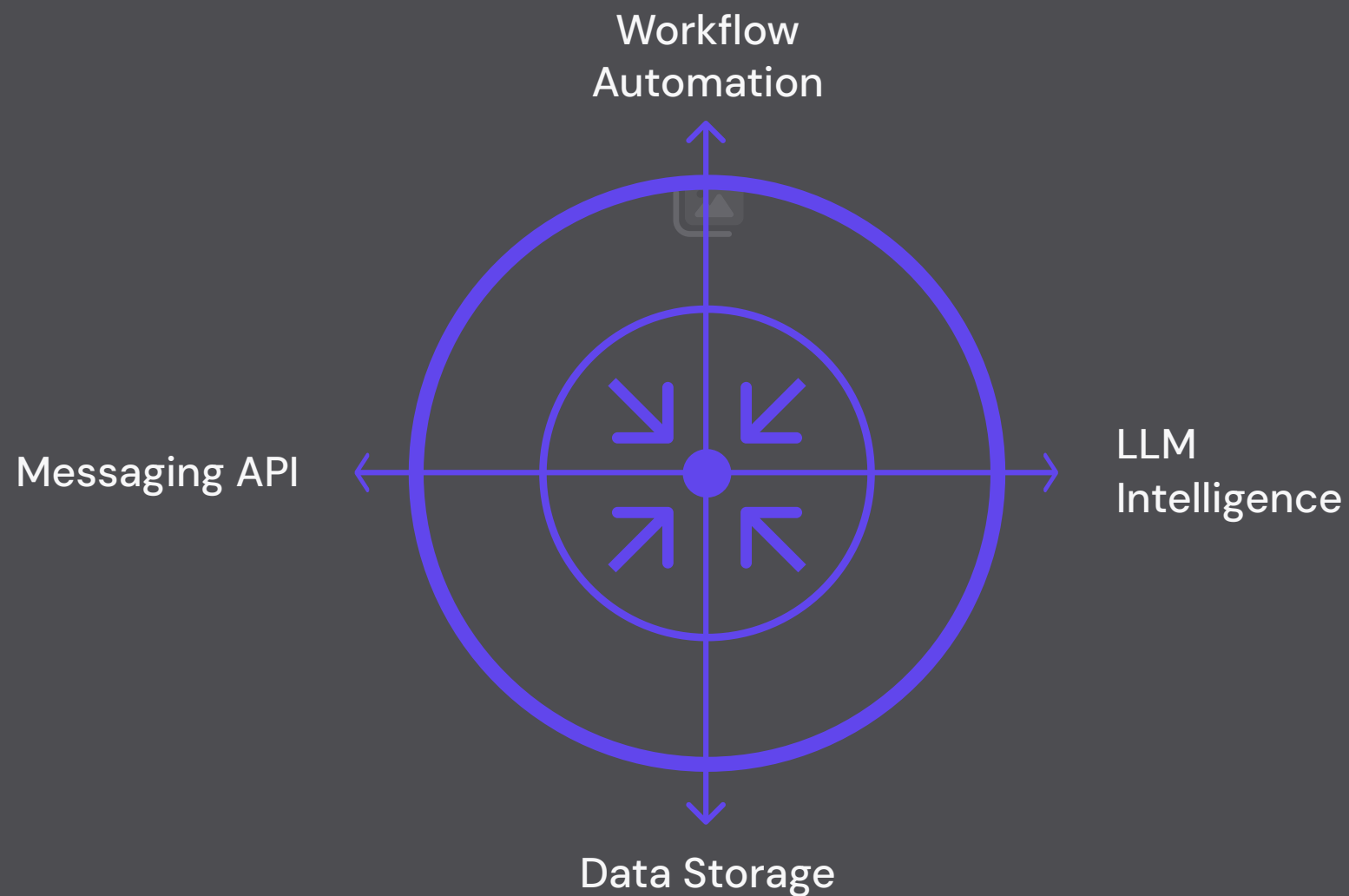


SupportFlow AI

Autonomous AI-Powered Customer Support Automation System

- n8n Workflow Automation
- Groq API + Llama LLMs
- Google Sheets Database
- Gmail API



The Problem: Traditional Support Bottlenecks

Manual Classification

No intelligent categorization — agents decide everything by hand

High Team Workload

Support staff overwhelmed with repetitive, low-value tasks

Slow Response Times

Hours or days to first reply — customers left waiting

No Automated Follow-Up

Tickets stall without progress or resolution checks

Incorrect Routing

Tickets sent to wrong departments, causing rework and frustration

Poor CX

Customer frustration from delays and systematic mishandling

Traditional

Customer → Human Agent → Manual Processing

SupportFlow AI

Customer → AI Automation → Instant Response

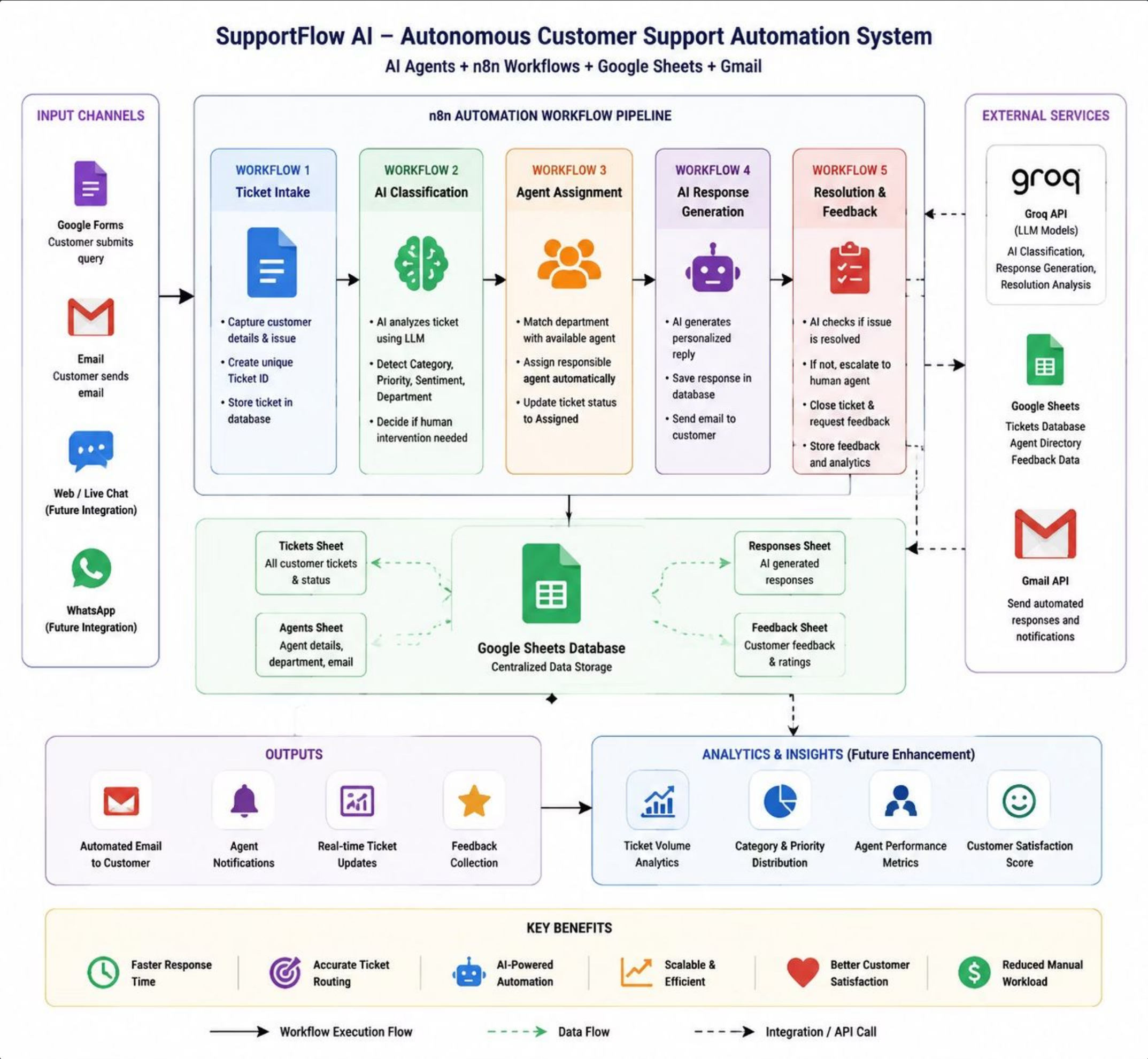
Solution: Intelligent Automation Pipeline

SupportFlow AI automates the complete customer support lifecycle — from intake to feedback — with zero manual intervention.



Result: Faster resolution, reduced manual work, improved customer satisfaction

System Architecture Overview



Key Components: Input channels (Forms, Email) → n8n automation engine → AI agents (classification, routing, response, resolution) → Google Sheets database → Gmail communication → Analytics & feedback

The diagram shows how SupportFlow AI connects every stage of the support workflow. Requests enter through Forms or Email, n8n orchestrates the automation, AI agents handle classification, routing, response, and resolution, and the system logs data in Google Sheets while communicating through Gmail and collecting analytics and feedback.

Workflow 1: Ticket Intake & Creation

Purpose: Collect customer issues and create structured support tickets with unique IDs



Flow: Customer → Google Form → n8n Trigger → Google Sheets
→ Ticket Record

1 Google Form Trigger

Collects customer name, email, subject, and issue description

Example input: "I was charged twice for my order"

2 Google Sheets — Add Ticket

Creates `ticket_id: TICKET-1786189064209`

Sets `status: New` and stores all customer details for downstream AI processing

Workflow 2: AI Ticket Classification

Purpose: Analyze new support tickets with Groq + Llama, extract structured fields, and route each case for the right action



Node 1: Google Sheets Read

Reads the newly created ticket record from Google Sheets and passes the full customer issue to the AI agent.

Input	Output
Ticket ID, customer name, email, subject, issue description	Structured ticket payload ready for LLM analysis

Node 2: Groq AI Agent

The AI analyzes the issue text and returns a classification result for downstream routing and escalation.

Category	Priority
Billing / Technical / Account / Shipping	High / Medium / Low
Sentiment	Routing
Positive / Neutral / Negative	Department assignment plus human escalation flag

AI Classification Outputs

- **Category:** Billing, Technical, Account, Shipping
- **Priority:** High, Medium, Low
- **Sentiment:** Positive, Neutral, Negative
- **Department:** Routed to the correct support team
- **Human Required:** True/False escalation flag

Example JSON Output

```
{
  "category": "Billing",
  "priority": "High",
  "sentiment": "Negative",
  "department": "Billing",
  "human_required": true
}
```

Flow: New Ticket → Google Sheets Read → Groq AI Agent → JSON Parser → Decision Node

Workflow 3: Automatic Agent Assignment

Purpose: Route classified tickets to the correct available support agent — automatically, with no human dispatcher needed

Assignment Pipeline



This flow routes each classified ticket to an available agent in the matching department, then updates the ticket automatically.

Node Details

Department Extraction

Reads department from AI classification output (e.g., "Billing").

Google Sheets Lookup

Searches Agents Sheet for available team members in the target department.

Example: Billing
Department → Agent:
Vidhaan
(agent@gmail.com)

Code Node

Formats the assignment payload before updating the ticket.

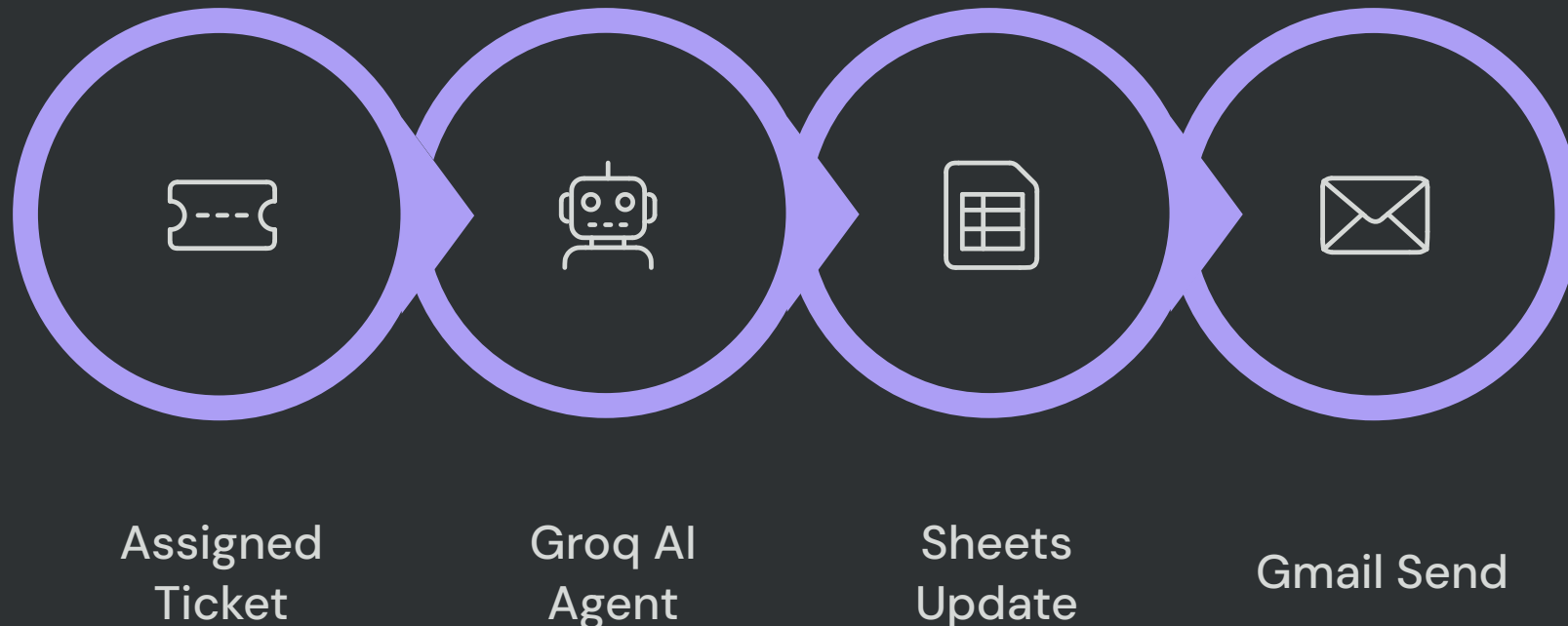
Update Ticket

Sets `assigned_agent = Vidhaan`, `status = Assigned`.

Workflow 4: AI Response Generation

Purpose: Generate personalized, professional customer responses automatically using Groq AI

Flow: Assigned Ticket → Groq AI Agent → Google Sheets Update → Gmail Send



Example response: "Hello Rahul, we received your billing complaint. Our team is reviewing the duplicate payment. You'll receive an update soon."

1 Retrieve Assigned Ticket

Fetches customer details, issue description, AI classification, and assigned agent from Google Sheets

2 AI Response Agent

Generates empathetic, professional responses explaining issue status and next steps — maintaining consistent support tone

3 Update & Send via Gmail

Stores `ai_response` in database, sets `status = Response Sent`, dispatches email via Gmail API

Workflow 5: Resolution & Feedback

Purpose: Verify ticket resolution and close the support cycle — automatically or escalate to humans when needed



TRUE — Resolved Path

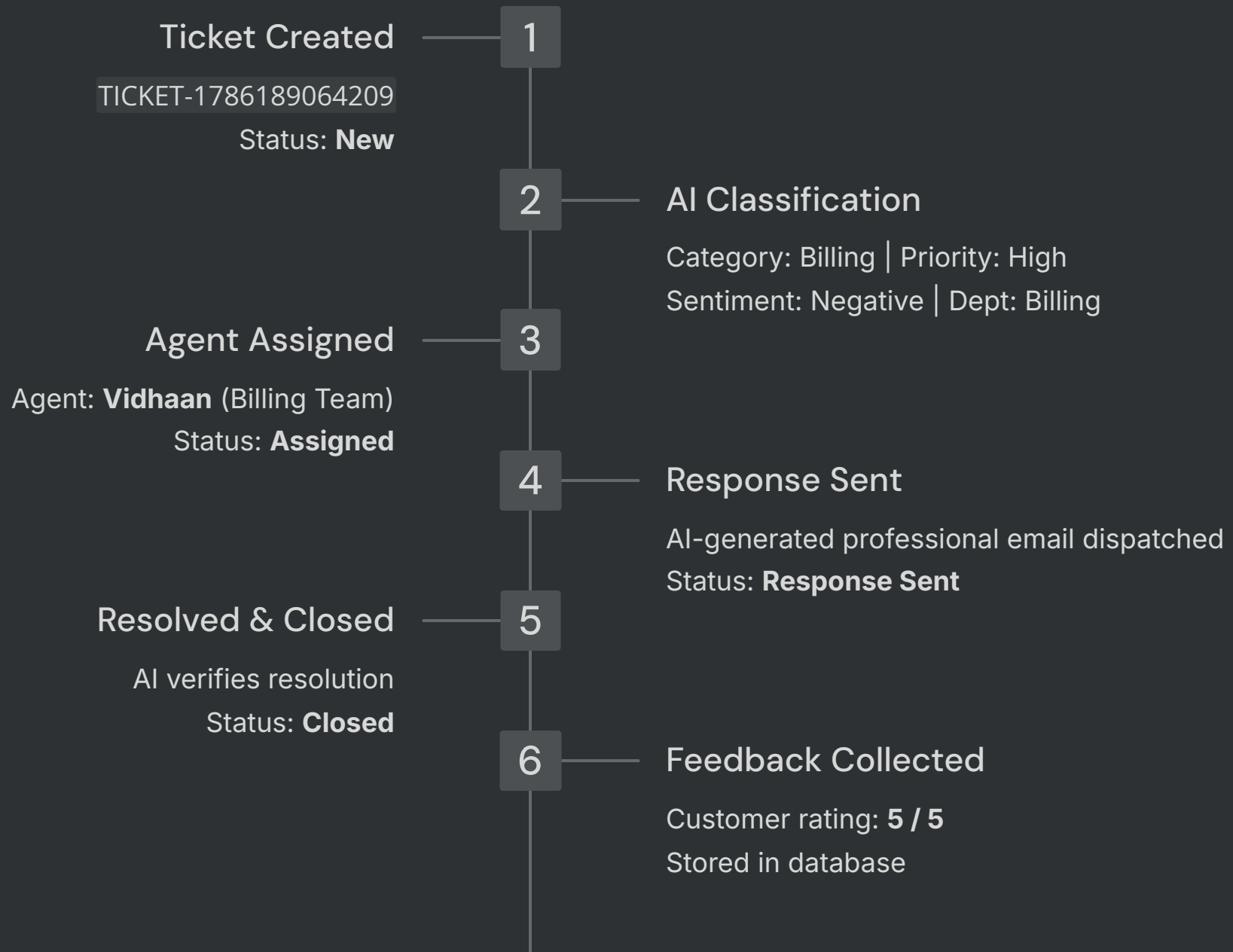
- Close ticket with timestamp
- Set `status = Closed`, record `closed_at`
- Send feedback request: *"Please rate your experience 1–5"*
- Store rating, comments & ticket reference

FALSE — Unresolved Path

- Mark `human_required = true`
- Notify support team for manual intervention
- Ticket stays active for follow-up

Complete End-to-End Example

Tracing a single customer issue — **"I was charged twice for my order"** — through all five workflows:



Technology Stack & Implementation



n8n — Automation Engine

Visual workflow builder orchestrating all data flow. Handles scheduling, triggers, and inter-system communication across the entire automation pipeline.



Groq API + Llama Models

Fast LLM inference for real-time classification and response generation. Llama provides advanced reasoning for intelligent routing, analysis, and automated replies.



Google Sheets — Database

Lightweight, API-accessible storage for tickets, agents, and feedback. Easy to query, update, and sync across all workflows.



Gmail API + Google Forms

Gmail API sends automated responses and notifications. Google Forms collects structured customer submissions at intake for seamless capture.



JavaScript Code Nodes

Custom transformation and decision logic embedded within n8n workflows for data formatting, enrichment, and conditional routing.